When Does Post-Hoc K=V Tying Fail?
A Convex-Hull Constraint in Self-Attention

Companion note to *Do Transformers Need Three Projections?* (Kayyam et al.)

Abstract

Do Transformers Need Three Projections? [1] shows that a Q≠K=V attention variant, trained *from scratch*, matches full QKV attention on a range of synthetic and vision tasks. This note asks a different question: what happens if K and V are tied *after* training, by surgically merging an already-trained QKV model's independent key and value projections, with no retraining? The central claim is narrow and architecture-level, not a claim about overall transformer expressivity: once $K = V$, a *single attention sub-block's* output is confined to the convex hull of the shared key/value vectors, regardless of the attention weights (Proposition 1); the residual stream, MLPs, and later layers are unrestricted by this and can and do carry information outside that hull. This predicts, and we confirm across five synthetic sequence tasks and four vision-classification datasets, a sharp split: the two synthetic tasks solvable without cross-token content routing are completely unaffected by the surgery, while every task or dataset that requires moving one position's value to another collapses toward chance. We then test the natural mechanistic story for the surgery-dependent cases – that the teacher's K-projection specialized for *addressing* at the expense of *content*, while V specialized the other way – and find it is only half right: a linear probe recovers token identity from K and V activations equally well, so the failure is not a loss of content information in K. A systematic interpolation between K and V, together with a random-projection control, shows that reusing K (KEEP_K) is often no better than an unrelated, untrained projection despite retaining linearly decodable token information – pointing to a representation-compatibility account rather than a simple information-loss one. Two further zero-shot experiments test this directly rather than infer it: decomposing attention into separate addressing- and payload-source projections shows payload identity, not addressing identity, drives the damage on several tasks, and – more decisively – a linear or orthogonal map from K to V's basis, fit on held-out data with no retraining of anything downstream, closes the entire KEEP_K gap and matches or exceeds KEEP_V on every task where KEEP_V itself recovers well, including the 300M-parameter language model. This is a direct manipulation of the hypothesized mechanism rather than inference from a probe and a random baseline, and we treat it as the strongest evidence in the note for the basis-mismatch account, though we still stop short of calling it an established mechanism, since a learned correction working is consistent with basis mismatch without ruling out every alternative account of the same result. Collapse severity also correlates with how functionally redundant the trained K/V projections already were (linear CKA on held-out activations, $r = -0.84$, $n = 9$, though this drops to $\rho = -0.56$ (n.s.) under Spearman and is not separately significant within either task-type group at $n = 5/n = 4$, so we treat CKA as a useful diagnostic rather than a validated general predictor) – and we surface, without resolving, a puzzle inside even that: within vision, K–V similarity *rises* rather than falls as classification difficulty increases from MNIST to CIFAR-100, even as collapse severity rises too. Finally, the collapse is comparatively cheap to repair: knowledge distillation from the original QKV model recovers the from-scratch tied baseline in 14 of 15 synthetic (task, surgery-mode) configurations within 5 epochs, and on vision, AVG-mode surgery reaches (and on CIFAR-10 slightly exceeds) the from-scratch tied baseline within 10 epochs, while KEEP_K remains the one mode with a persistent, slow-converging gap – consistent with the basis-mismatch account, since it is the mode with the most re-calibration to do. A single

confirmatory run at a scale closer to production pretraining – a 300M-parameter language model trained on 10B tokens of FineWeb-Edu – replicates the same AVG > KEEP_V > KEEP_K ordering in both zero-shot severity and post-distillation gap, and a cheap K/V-redundancy check (CKA) extends to that scale too. A content probe complicates the small-scale picture, though: unlike the synthetic/vision teachers, where K and V decoded token identity equally well, this teacher's V probes consistently better than K across all 20 layers (a modest but universal gap), closer to Corollary 1's original content-loss story than the basis-mismatch account we favor at small scale. We did not repeat the random-c_{kv} baseline or α-sweep at that scale. Our overarching point is not that $K = V$ is fundamentally deficient: jointly training `Q≠K=V` from scratch works fine (that is the base paper's headline result), and untying is trivial for the two tasks that never needed cross-token routing. The point is that *post-hoc merging of independently-trained representations is a different operation from a joint architectural constraint*, with its own, separately characterizable failure mode.

1 Setup and notation

Relationship to the base paper. *Do Transformers Need Three Projections?* asks whether `Q≠K=V`, trained jointly from a random initialization, can match full `QKV` attention – and finds that it can. This note asks a different question about the same equality: can the representations learned by *independent* K and V projections, already specialized apart over a full training run, be merged *after the fact*? These are not the same intervention. Joint training lets addressing and content pressures negotiate a single shared representation from the start, under gradient descent's full control; post-hoc merging takes two representations optimized under no such constraint and imposes it externally, with no opportunity for the rest of the network to adapt. The central scientific question here is specifically the latter: whether, and how predictably, independently-trained representations resist being merged – not whether $K = V$ is expressive enough when it is a design choice made before training starts.

We use the attention parameterization of the base paper: for input $x \in \mathbb{R}^{T\times d}$, single-head[1] self-attention computes

$$q_i = c_q(x_i), \quad k_j = c_k(x_j), \quad v_j = c_v(x_j), \quad a_{ij} = \mathrm{softmax}_j\Big(\frac{q_i \cdot k_j}{\sqrt{d}}\Big), \quad y_i = \sum_{j=1}^{T} a_{ij}\, v_j,$$

where c_q, c_k, c_v are learned linear maps. The base paper's six variants are recovered by sharing these maps: `QKV` (none shared), `Q=K≠V` ($c_q = c_k$), `Q≠K=V` ($c_k = c_v$), and `Q=K=V` (all three shared), each optionally with a 2D positional injection (the "+" variants). Every block also carries a residual connection, $h_i^{\mathrm{out}} = h_i^{\mathrm{in}} + \mathrm{Attn}(\cdot)_i$, followed by a position-wise MLP with its own residual connection.

Surgery. Given a *trained* `QKV` checkpoint (independent c_k, c_v), we construct a `Q≠K=V` student with no retraining by setting the student's shared projection c_{kv} to one of:

- KEEP_K: $c_{kv} \leftarrow c_k$ (drop the trained V entirely),
- KEEP_V: $c_{kv} \leftarrow c_v$ (drop the trained K entirely),
- AVG: $c_{kv} \leftarrow \frac{1}{2}(c_k + c_v)$ (average the two Linear layers' weights and biases; since both are affine maps this is identical to averaging their outputs).

All other parameters (c_q, c_{proj}, MLPs, layer norms, embeddings, head) are copied verbatim from the teacher.

[1] The multi-head case is identical after splitting d into H blocks of size d/H; the argument in Section 2 applies unchanged within each head.

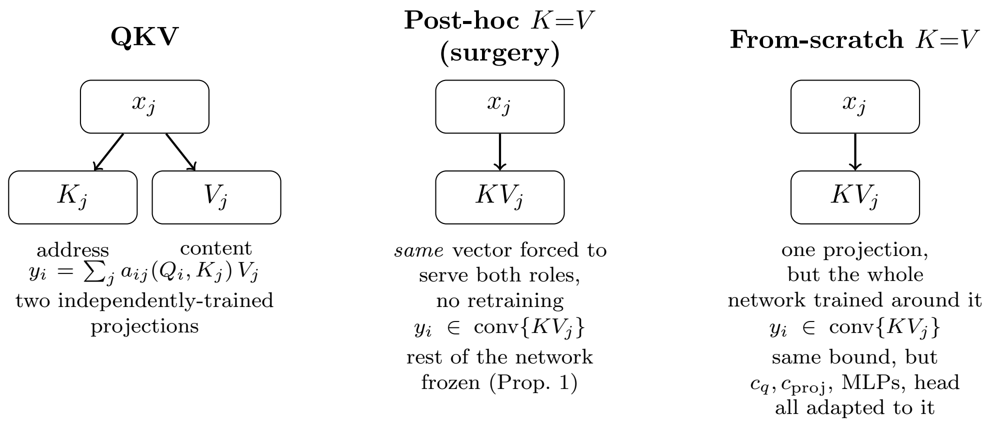

Figure 1: The same architecture, K=V, reached two different ways. Left: the QKV teacher, with independent K/V projections free to specialize for addressing and content separately. Middle: post-hoc surgery forces one already-specialized representation to do both jobs, with nothing else in the network adjusted (Section 4.2 finds the result closer to a basis mismatch than to information loss). Right: choosing K=V *before* training lets every other parameter co-adapt to it from the start – the same mathematical bound (Proposition 1) applies, but nothing has to unlearn a different assumption.

2 A convex-hull obstruction

Proposition 1 (Convex-hull confinement under K=V)**.** *If $k_j = v_j$ for every position j (as holds for all three surgery modes above, and for the from-scratch* `Q=K=V` *and* `Q≠K=V` *variants), then for every query position i,*

$$y_i \in \mathrm{conv}\{k_1, \ldots, k_T\} \subseteq \mathrm{span}\{k_1, \ldots, k_T\},$$

regardless of the attention weights a_{ij}.

Proof. $a_{ij} \geq 0$ and $\sum_j a_{ij} = 1$ by construction of the softmax, so $y_i = \sum_j a_{ij} k_j$ is by definition a convex combination of $\{k_j\}$. □

Corollary 1. *The attention sub-block can only make available to position i information that is already representable inside* span$\{k_1, \ldots, k_T\}$*. If the trained K-projection was shaped by gradient descent to produce a good* addressing *signal ($q_i \cdot k_j$ ranking the correct target highest) without pressure to remain* injective *in the token content x_j — because that pressure was carried entirely by the independent V-projection in the QKV teacher — then k_j need not determine x_j, and routing k_j itself to position i can lose the very content the downstream task needs. This is a hypothesis about* why *surgery hurts, stated here before we test it; Section 4.2 probes addressing and content recoverability directly and finds the picture is more specific than "K lacks content" as stated – content is present in K too, but evidently in a form the rest of the network cannot use.*

Remark 1 (Invariance to normalization)**.** *Proposition 1's proof uses only that $a_{ij} \geq 0$ and $\sum_j a_{ij} = 1$; it places no assumption on how k_j (equivalently v_j, since $k_j = v_j$) is computed. It therefore holds unchanged whether k_j is a raw linear projection, an ℓ_2-normalized projection (as in QK-normalized attention variants), or the output of any other fixed transformation applied before the weighted sum – confinement is to the convex hull of whatever vectors are actually summed, in whatever geometry they live in. In the checkpoints studied here, LayerNorm is applied pre-block to* *the residual stream x_j before c_k, c_v project it (Section 1), not to k_j or v_j themselves, so this is a non-issue for our experiments specifically; we state the general invariance here because it is easy to construct architectures where it would otherwise look like a threat to the argument. A normalization that discarded information v_j needed to retain would show up as a smaller effective* $\operatorname{span}\{k_j\}$ *in Corollary 1's premise, not as a failure of Proposition 1 itself, which is unconditional.*

Remark 2 (Scope: one attention sub-block, not the whole transformer)**.** *Proposition 1 constrains y_i, the output of one attention sub-block. It does not constrain $h_i^{out} = h_i^{in} + y_i$, the MLP applied to it, or any later layer: the residual stream carries h_i^{in} forward regardless of what the attention sub-block computes, and nothing here restricts what a downstream MLP or a later attention layer (with its own, unconstrained K'/V') can do with the result. A model with $K = V$ in every layer is therefore* not *restricted to representing only what lies in a single layer's convex hull – claiming that would be a claim about the whole network's expressivity, which Proposition 1 does not make and we do not intend. What the proposition does say is narrower and, we think, still useful: a single $K = V$ attention operation cannot independently choose the representation used for content transport from the representation used for addressing – what gets transported to position i must be a convex combination of the very vectors $\{k_j\}$ that determined* where *to look, not some separately-chosen payload. Whether that narrower constraint ends up mattering for a given task is exactly the empirical question the rest of the note is about.*

Proposition 1 is therefore silent on *when* its obstruction actually bites in practice: a task that never needs cross-token content routing in the first place is immune to it regardless of how many layers have $K = V$. Section 3 makes this precise for our tasks; Section 4 shows that even among the routing-dependent tasks, the practical severity varies with how much the teacher's K and V projections had actually specialized (Corollary 1's premise) – which we measure and test directly rather than assume, and which turns out to need a more specific account than the corollary as originally stated (Section 4.2).

3 Why pointwise tasks are immune

Two of the five synthetic tasks are pointwise: SUB computes $y_i = 9 - x_i$ and COPY computes $y_i = x_i$, each a function of position i's own input only. Each block's residual connection preserves a direct, additive path from position i's own incoming representation ($\text{embed}(x_i) + \text{pos\_emb}(i)$ at the first block, that block's own input thereafter) through to its output, regardless of what the attention sub-block computes; stacked over every block, that representation is never overwritten, only added to. Combined with per-position MLPs that are already universal approximators of pointwise maps, attention's specific content-routing is therefore not load-bearing for these two tasks, and Proposition 1's obstruction is vacuous: it restricts what attention can contribute, but the task does not need attention to contribute anything.

This is not merely a plausibility argument – it is directly testable with data we already have. If attention's specific query–key matching mattered for SUB/ COPY, corrupting the value stream with either the wrong projection (KEEP_K) or corrupting the key stream with the wrong projection (KEEP_V) should hurt *at least one* of the two directions. Instead, Table 1 shows *both* surgeries leave SUB and COPY at 1.000 accuracy, exactly as the residual-bypass account predicts, while the three routing-dependent tasks (REVERSE, SORT, SWAP – each requires moving a *different* position's value to position i) collapse sharply under the same surgeries.

4 Empirical validation

4.1 Zero-shot surgery and distillation recovery

Table 1 and Table 2 report, for each task/dataset: the from-scratch `QKV` teacher accuracy, the from-scratch tied (`Q≠K=V`) baseline accuracy (a reference performance a distilled student can reasonably target – not a hard ceiling, since AVG/CIFAR-10 slightly exceeds it (see Table 2)), the zero-shot accuracy immediately after surgery (no fine-tuning), and the accuracy after 5 epochs of distillation from the QKV teacher (cross-entropy on hard labels + KL divergence against the teacher's soft logits, $T = 2$, equal weighting).

Task	Mode	QKV	Q≠K=V (scratch)	Zero-shot surgery	Distilled (5 ep)
REVERSE	keep_k	1.000	1.000	0.100	1.000
	keep_v	1.000	1.000	0.123	1.000
	avg	1.000	1.000	0.861	1.000
SORT	keep_k	0.998	0.996	0.583	0.965
	keep_v	0.998	0.996	0.717	0.996
	avg	0.998	0.996	0.751	0.996
SUB	any	1.000	1.000	1.000	1.000
SWAP	keep_k	1.000	1.000	0.088	1.000
	keep_v	1.000	1.000	0.122	1.000
	avg	1.000	1.000	0.906	1.000
COPY	any	1.000	1.000	1.000	1.000

Table 1: Synthetic sequence tasks (token accuracy). Routing-dependent tasks (REVERSE, SORT, SWAP) collapse under surgery and recover under distillation; pointwise tasks (SUB, COPY) are unaffected by either. 14 of these 15 (task, surgery-mode) configurations reach the from-scratch tied baseline within 5 epochs; the one exception, SORT/KEEP_K, reaches 0.965 against a 0.996 target and is still improving at epoch 5.

AVG is uniformly the safer zero-shot surgery on every vision dataset, and doubling the distillation budget does not treat all three surgery modes equally: KEEP_K is the one mode with a persistent gap even at 10 epochs (Section 4.2 tests why, on the synthetic tasks, and finds the evidence more consistent with a basis-mismatch account than a simple content-loss one).

Tables 1–2 report only endpoint accuracies; Figure 2 shows the full per-epoch recovery curve for every (task or dataset, surgery-mode) configuration referenced above. The shape confirms what the endpoints suggest: REVERSE/SUB/SWAP/COPY and MNIST are already flat by epoch 1 (the surgery was mild or the gap closes almost immediately), SORT/KEEP_K and every CIFAR/KEEP_K curve are still rising through their full epoch budget without having visibly plateaued as early as epoch 2–3, and KEEP_V/AVG on CIFAR flatten out well before their budget ends – consistent with KEEP_K being a harder optimization problem rather than merely a deeper hole to climb out of.

Where does the zero-shot optimum sit, and is reusing K better than noise? Tables 1–2 report only three points along the line $c_{kv} = \alpha W_K + (1 - \alpha)W_V$ (α=1: KEEP_K, α=0: KEEP_V, α=0.5: AVG). Figure 3 sweeps $\alpha \in \{0, 0.1, \ldots, 1.0\}$ and adds a fourth reference point: c_{kv} left at its fresh, untrained initialization, with no relationship to the teacher's K or V at all – reported as a mean ± 1 std. over 5 random seeds, not a single number, since the baseline's own sampling variance turns out to matter for what we can claim about it.

Two things stand out. First, the curve is neither monotonic nor symmetric around AVG: REVERSE, SWAP, and all four vision datasets peak near $\alpha \approx$ 0.4–0.5, while SORT peaks earlier, around $\alpha \approx$ 0.2– 0.3, and degrades monotonically from there to KEEP_K. SUB and COPY are flat

Dataset	Mode	QKV	Q≠K=V (scratch)	Zero-shot surgery	Distilled (5 ep)	Distilled (10 ep)
MNIST	keep_k	0.981	0.978	0.114	0.979	–
	keep_v	0.981	0.978	0.151	0.982	–
	avg	0.981	0.978	0.910	0.983	–
FMNIST	keep_k	0.887	0.882	0.084	0.881	–
	keep_v	0.887	0.882	0.105	0.883	–
	avg	0.887	0.882	0.811	0.888	–
CIFAR-10	keep_k	0.696	0.698	0.076	0.631	0.666
	keep_v	0.696	0.698	0.153	0.686	0.693
	avg	0.696	0.698	0.368	0.696	**0.702**
CIFAR-100	keep_k	0.437	0.445	0.007	0.384	0.411
	keep_v	0.437	0.445	0.021	0.424	0.430
	avg	0.437	0.445	0.109	0.432	0.435

Table 2: Vision classification (ViT, top-1 accuracy). Unlike the synthetic tasks, every dataset shows some collapse under KEEP_K/KEEP_V: these ViTs classify from a learned CLS token (`h[:, 0]` after the final block), not global average pooling, so which patches reach the classification decision is itself an attention-weighted, content-routing operation rather than a fixed uniform average – there is no pointwise bypass analogous to Section 3. At the 5-epoch budget, KEEP_K on CIFAR-10 and CIFAR-100 was still visibly improving, so both CIFAR datasets (all three modes) were rerun with a 10-epoch budget; MNIST/FMNIST were not rerun since they were already converged by 5 epochs. KEEP_V and AVG are essentially converged on CIFAR by 10 epochs (AVG/CIFAR-10 slightly exceeds the from-scratch tied baseline); KEEP_K closes only about half its remaining gap, and its per-epoch curve is already plateauing by epoch 8–9 (full curves in `distillation_vision_cifar{10,100}_10ep.csv`), indicating a genuinely harder, slower-converging recovery rather than simple under-training at 5 epochs.

at 1.000 across every α, including every random seed – a stronger confirmation of Section 3 than the three original discrete points.

Second: KEEP_K does not reliably preserve useful downstream functionality despite preserving linearly decodable content (Section 4.2). Table 3 compares KEEP_K against the random baseline's mean and std. for each non-pointwise task. On SWAP, KEEP_K sits about 12 random- baseline standard deviations below the mean – unambiguous. On the other five (REVERSE, SORT, MNIST, FMNIST, CIFAR-10, CIFAR-100), KEEP_K sits within about 2 standard deviations of the random baseline in either direction, which at $n = 5$ seeds we cannot call a reliable difference from noise in either direction. We had originally read the point estimates alone as "KEEP_K is usually at or below random," which overstated what 5 seeds actually support; the defensible claim is narrower and, we think, still the interesting one: outside of SWAP's dramatic case, KEEP_K's specialized (non-random) K-derived representation confers no detectable advantage over uninformative noise to the frozen downstream network, even though Section 4.2's probe shows that representation is just as linearly content-rich as V's.

Confirmation at language-model scale. Everything above uses small models (1–2M-param vision ViTs, tiny sequence models) where a full α-sweep and a 5-seed random baseline are cheap. To check the collapse-and-recover pattern at a scale closer to production LLM pretraining, we repeat the core protocol – QKV teacher, surgical tying, zero-shot eval, distillation recovery – on the base paper's 300M-parameter GPT-style model, trained from scratch on 10B tokens of FineWeb-Edu (`outputs_qkv_baseline_10B/final_model.pt`, validation perplexity 20.82), against the base paper's matching from-scratch `Q≠K=V` checkpoint (perplexity 21.28) as the tied ceiling. Distillation

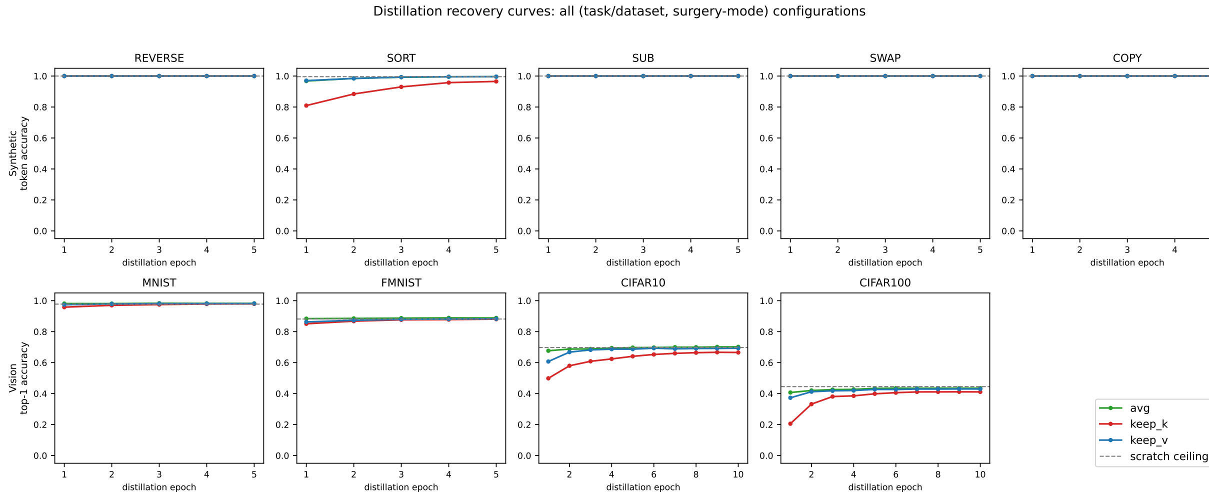

Figure 2: Per-epoch distillation accuracy for every (task/dataset, surgery-mode) configuration in Tables 1–2 (dashed line: the from-scratch Q≠K=V tied baseline). CIFAR-10/CIFAR-100 panels show the full 10-epoch runs; all other panels are the 5-epoch runs (not rerun longer since already converged).

Task/dataset	KEEP_K	random c_{kv} (mean $\pm$ std, 5 seeds)	z
REVERSE	0.100	0.100 ± 0.0004	$+1.25$
SORT	0.583	0.584 ± 0.0099	-0.12
SWAP	0.088	0.100 ± 0.0010	-12.30
MNIST	0.114	0.091 ± 0.0124	$+1.92$
FMNIST	0.084	0.118 ± 0.0278	-1.23
CIFAR-10	0.076	0.112 ± 0.0210	-1.72
CIFAR-100	0.007	0.011 ± 0.0024	-1.88

Table 3: KEEP_K vs. the random-c_{kv} baseline, with z = (keep_k − mean)/std. Only SWAP is a large, unambiguous effect; the rest are within about 2 standard deviations of the random baseline's own spread at $n = 5$ seeds, i.e. not reliably distinguishable from noise in either direction. SUB and COPY are omitted: every mode, including all 5 random seeds, scores exactly 1.000.

uses the same hard-label-CE + KL-vs.-teacher-soft-logits objective (T=2, equal weighting) as Tables 1–2, but for a 500M-token budget (about 5% of the from-scratch pretraining budget, in the same spirit as the epoch budgets used above) rather than a fixed epoch count, since "epoch" is not a meaningful unit against a pretraining corpus far larger than the distillation budget itself.

The ranking replicates exactly. AVG is again the safest zero-shot surgery and recovers closest to the tied ceiling (22.87 vs. 21.28, a 1.6-point gap after 500M distillation tokens), echoing AVG/CIFAR-10's near-complete recovery in Table 2; KEEP_K is again both the worst zero-shot mode and the one with the largest residual gap after distillation (29.00 vs. 21.28), matching the persistent KEEP_K gap already seen at vision scale (Section 5). The teacher's K/V activation CKA – cheap to measure, since it needs only a forward pass on held-out data and no retraining – is 0.70, squarely inside the range spanned by the 9 synthetic/vision teachers. Converting perplexity to a chance-normalized log-loss severity (Figure 4's caption gives the exact transform) places the LLM point within the same scatter around Figure 4's $n = 9$ fit line as the other 9 points (residual 0.19, in the same range as, e.g., FMNIST's 0.14 or REVERSE's 0.25) despite not being part of it – consistent with, though

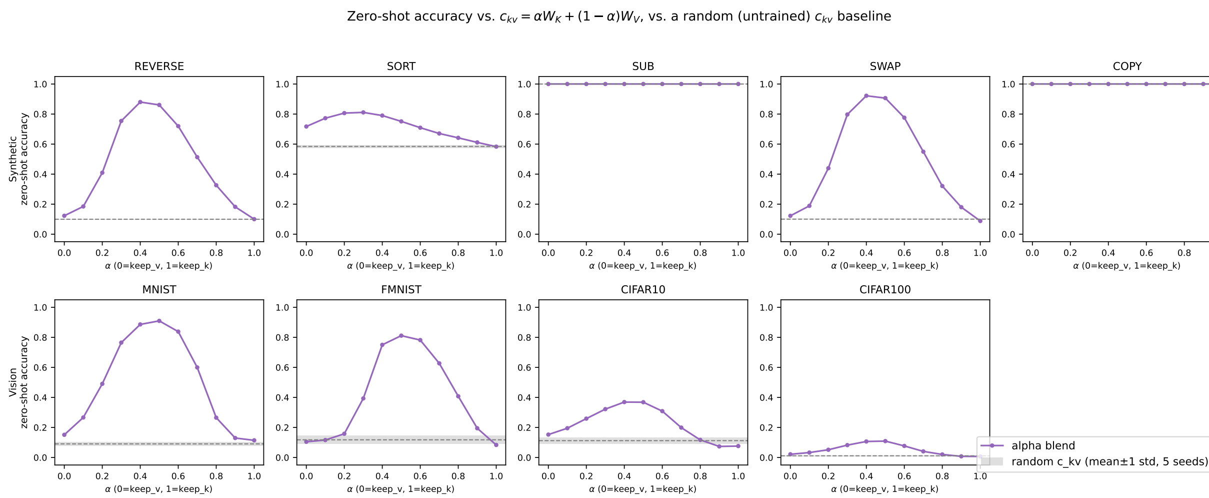

Figure 3: Zero-shot accuracy as c_{kv} is swept from KEEP_V (α=0) to KEEP_K (α=1), vs. a random-initialization c_{kv} baseline (shaded: mean $\pm$ 1 std. over 5 seeds). No retraining anywhere in this figure.

not a tight confirmation of, the CKA–severity relationship extending to a domain and scale neither of the original 9 points covers. We do not fold it into that figure's $r = -0.84$ correlation regardless, since the normalization is a different derivation from the other 9 points' raw accuracy drop, not a validated equivalent measure.

A cheap, no-retraining content probe (same closed-form ridge-regression method as Section 4.2, adapted for the 50,304-token vocabulary via an index-add accumulation in place of a dense one-hot target matrix, and with the ridge penalty scaled to the data's own diagonal energy rather than a fixed constant – this model's K/V activations turn out to be severely rank-deficient, and a fixed ridge silently degenerates the closed-form solve into float32 noise at one layer if left unscaled) tells a different story from the synthetic/vision result, though. Averaged over the model's 20 layers, the probe recovers token identity from v_j noticeably better than from k_j (85.3% vs. 80.3% mean accuracy, a 5.0-point gap on average, ranging from 1.7 to 10.9 points across layers, with V ahead at every single one – full per-layer numbers in `kv_content_probe_llm_results.csv`). At synthetic/vision scale, K and V probed equally well, which is precisely what motivated moving away from Corollary 1's original content-loss story toward the basis-mismatch account (Section 4.2). Here, the LLM teacher's K *is* measurably harder to decode token identity from than V – closer to Corollary 1 as originally stated. We read this as a genuine point of divergence rather than force it into the existing account: the gap is consistent in direction across every layer, but modest in size (both projections still probe well above 80%), so it does not on its own settle whether basis-mismatch, partial content-loss, or both is the better description at this scale – and, like the other single-point checks in this section, it comes from one held-out batch, not a bootstrapped or multi-seed estimate.

We did not repeat the α-sweep or random-c_{kv} baseline (Section 4.2 onward) at this scale – each would cost a multi-GPU-day training run per point here rather than the few seconds it takes at vision/synthetic scale – so this result is best read as a single confirmatory data point on the headline collapse-and-recover pattern, not a full replication of the mechanistic analysis.

Mode	QKV teacher	Q≠K=V (scratch)	Zero-shot surgery	Distilled (500M tok)
keep_k	20.82	21.28	13345.19	29.00
keep_v	20.82	21.28	7347.14	27.00
avg	20.82	21.28	6430.36	22.87

Table 4: FineWeb-Edu language modeling (300M params, validation perplexity – lower is better, unlike the accuracy tables above). Causal next-token prediction is not a pointwise task in the sense of Section 3: every position's prediction depends on routing information forward from earlier tokens, so Proposition 1 predicts collapse under $K = V$ tying, and zero-shot surgery confirms it dramatically – perplexity worsens by two to three orders of magnitude (20.8 → 6,400–13,300) versus the roughly one-order-of-magnitude accuracy collapses in Tables 1–2, consistent with perplexity's exponential sensitivity to per-token log-loss rather than a qualitatively different failure mode. Distillation recovers all three modes to within 1.6–7.7 perplexity points of the scratch-tied ceiling (AVG closest, KEEP_K furthest), reproducing the same AVG > KEEP_V > KEEP_K ordering, by both zero-shot severity and post-distillation gap, seen in Table 2.

4.2 Testing the addressing/content mechanism directly

Corollary 1 rests on an assumption we had not, until now, tested directly: that the teacher's K-projection specialized for *addressing* at some cost to *content*, while V specialized the other way. We test both halves on the five synthetic-task teachers, where the correct routing target at each position is known exactly (COPY/SUB: self; REVERSE: position $T-1-i$; SWAP: swap the two halves; SORT: the source position is $\arg\mathrm{sort}(x)$, which is per-sample and data-dependent).

Addressing. For each layer, we take the model's actual attention weights a_{ij} (re-derived from the true layer-normed block input via a forward hook, not recomputed independently) and check whether $\arg\max_j a_{ij}$ matches the task's ground-truth source position. On the two genuinely one-to-one permutation tasks this is strong (REVERSE 1.00 → 0.83 from layer 0 to 1; SWAP 1.00 → 0.78). On SUB and COPY it is near chance (≈ 0.02–0.05) at both layers, consistent with Section 3: attention apparently does not implement self-addressing for these, because it does not need to. SORT is *also* near chance under this diagnostic, which we do not read as evidence against SORT needing routing – it collapses under surgery just like REVERSE/SWAP (Table 1) – but as a sign that "does argmax match one ground-truth index" is the wrong operationalization for a task whose natural solution is a distributed rank/comparison computation over many positions rather than a hard one-to-one copy. We report this as a limitation of the diagnostic, not a finding about SORT's mechanism.

Content. We fit a closed-form ridge-regression linear probe from k_j (resp. v_j) to x_j's one-hot identity, on a held-out train/test split, and report test-set classification accuracy. The result complicates Corollary 1 as stated: at this model's width (d=256), *both* k_j and v_j probe at or near 1.000 on every task and both layers, with no consistent gap (the one partial exception, SORT layer 1, is 1.000 for K vs. 0.991 for V – if anything the wrong direction for the hypothesis). So the teacher's K-projection has not, in any sense a linear probe can detect, sacrificed the ability to reconstruct token identity. Corollary 1's premise – that K is under no pressure to stay content-decodable – does not hold as a loss of *linearly available information*, at least not at this scale.

Read together with the alpha-sweep/random-baseline result above, this points to a more specific mechanism than "K lost the content": KEEP_K confers no detectable advantage over an untrained random baseline on most non-pointwise tasks, and is dramatically worse on one (SWAP; Table 3), even though k_j is linearly just as decodable as v_j. K does not reliably preserve useful downstream functionality despite preserving content, which is consistent with a **basis mismatch** rather than an information loss – though we have not measured downstream calibration to K vs. V directly, so we hold this as the best-supported reading rather than an established mechanism: k_j carries x_j's identity in a representation shaped by the addressing objective ($q_i \cdot k_j$ ranking correctly), not in the representation c_{proj}, the MLP, and the head were calibrated against – and handing those frozen downstream weights an unfamiliar (if information-rich) basis is, on the evidence in Table 3, no more useful to them than noise. We did not anticipate this when first writing Corollary 1, and we think it is a more precise and more interesting claim than the one we originally set out to test – and it is also consistent with why distillation recovers KEEP_K slowest (Section 5): re-calibrating a frozen network to a new, unfamiliar basis is a harder fitting problem than recovering from noise-level information loss would be.

4.3 Two direct tests of the basis-mismatch account

Everything above is indirect evidence for basis mismatch: a probe showing content survives, a random baseline showing that content is nonetheless useless, a distillation-speed asymmetry consistent with re-calibration being the bottleneck. None of it manipulates the hypothesized mechanism directly. We add two zero-shot (no retraining) experiments that do: separately vary which projection produces attention *scores* versus which produces the *transported* vectors, and directly correct K's basis toward V's with a learned map fit on held-out data.

Addressing/payload decomposition. Writing $A(S, P)$ for attention with scores from S and transported payload from P, the teacher computes $A(Q, K)V$ and there are three other cells: $A(Q, K)K$, $A(Q, V)V$, $A(Q, V)K$. Two of these are not new experiments – they are exactly what KEEP_K/KEEP_V already measured: tying $c_{kv} \leftarrow c_k$ reuses c_k for *both* roles, so scores are unchanged ($A(Q, K)$) and the payload becomes K; symmetrically for KEEP_V and c_v. Only $A(Q, V)K$ – scores from V, payload from K – is genuinely new. We evaluate it by monkey-patching the teacher's own attention forward (no new model, no retraining), on the same held-out data as Tables 1–2.

Task/dataset	$A(Q,K)V$ (teacher)	$A(Q,K)K$ (KEEP_K)	$A(Q,V)V$ (KEEP_V)	$A(Q,V)K$ (new)
REVERSE	1.000	0.100	0.123	0.104
SORT	0.998	0.583	0.717	0.576
SWAP	1.000	0.088	0.122	0.108
MNIST	0.981	0.114	0.151	0.081
FMNIST	0.887	0.084	0.105	0.099
CIFAR-10	0.696	0.076	0.153	0.073
CIFAR-100	0.437	0.007	0.021	0.006

Table 5: Addressing/payload decomposition, zero-shot accuracy (SUB/COPY omitted: 1.000 in every cell, as always). $A(Q, V)K$ tracks $A(Q, K)K$ (KEEP_K) closely on SORT and both CIFAR datasets – evidence that *payload* identity, not addressing identity, drives the damage there, directly supporting the basis-mismatch reading of KEEP_K's failure. REVERSE, SWAP, and MNIST instead collapse to a similar floor regardless of the combination, consistent with these one-to-one-routing tasks being sensitive to any deviation from the exact trained pairing.

At LLM scale the same swap gives val. PPL 9622.90 for $A(Q, V)K$ – between KEEP_K (13345.19) and KEEP_V (7347.14), mildly closer to KEEP_V, i.e. a less clean "payload dominates" signal than SORT/CIFAR show. We read this as a genuine, honestly-reported scale-dependence rather than force one story onto both: at small scale payload identity is often the dominant factor, but the LLM result does not cleanly replicate that.

K→V alignment. A more direct test: if KEEP_K's problem is really that c_{proj}/MLP/head are calibrated to V's basis, then a simple, *learned* change of basis from K to V – fit on held-out data, applied with no retraining of anything downstream – should recover most of the gap. We fit two such corrections per layer, on a held-out split disjoint from the accuracy-reporting test set: a ridge-regularized linear map $A^\star, b^\star = \arg\min_{A,b} \sum_j \|Ak_j + b - v_j\|^2$, and an orthogonal Procrustes rotation $R^\star = \arg\min_{R^\top R=I} \|KR - V\|_F$ (closed-form via SVD). Both replace KEEP_K's literal $c_{kv} \leftarrow c_k$ with $c_{kv}(x) \leftarrow A^\star c_k(x) + b^\star$ (resp. $c_k(x)R^\star$), evaluated zero-shot.

Task/dataset	KEEP_K	KEEP_K + linear	KEEP_K + Procrustes	KEEP_V
REVERSE	0.100	0.118	0.107	0.123
SORT	0.583	**0.740**	0.710	0.717
SWAP	0.088	0.118	0.107	0.122
MNIST	0.114	**0.151**	0.113	0.151
FMNIST	0.084	**0.109**	0.100	0.105
CIFAR-10	0.076	0.152	**0.170**	0.153
CIFAR-100	0.007	0.023	**0.025**	0.021

Table 6: Zero-shot accuracy after correcting KEEP_K's c_{kv} toward V's basis with a learned map, no retraining. Bold: matches or exceeds KEEP_V. On SORT and every vision dataset, either correction closes the entire KEEP_K gap and reaches or exceeds KEEP_V – direct evidence that the frozen downstream network can use K's content once it is expressed in a compatible basis, not indirect inference from a probe and a random baseline. REVERSE/SWAP recover only to KEEP_V's own (still low) level, consistent with those tasks needing distillation regardless of starting projection (Table 1).

At LLM scale, Procrustes gives val. PPL 9281.23 and the linear correction gives 6666.60 – both far below KEEP_K's 13345.19, and the linear correction in fact *beats* KEEP_V (7347.14), mirroring the small-scale pattern closely. Getting there took real debugging, worth reporting because it is itself informative: an unconstrained 1024×1024 map fit from a single ∼16k-token batch is underdetermined enough that a fixed ridge penalty tuned for the tiny synthetic/vision models (where d is small relative to sample size) catastrophically overfits here (val. PPL 38747 – *worse* than plain KEEP_K) unless the ridge strength is scaled up substantially. Worse, our first fix – selecting the ridge by reconstruction error ($\|Ak + b - v\|^2$) on a disjoint held-out batch of training documents – was itself wrong: that criterion monotonically favors the *worst*-performing ridge at every layer, because minimizing embedding-space reconstruction error is not the same objective as producing a representation the frozen downstream computation can use – the identical disconnect this section's content probe already found between K's linear decodability and its downstream uselessness, surfacing again one level up, in how to fit the correction rather than in the correction's target. Selecting the ridge by actual downstream perplexity on a held-out (non-reported) split, rather than embedding reconstruction error, resolves it. We do not read the LLM result as less trustworthy for having needed this fix – if anything, needing to avoid the exact failure mode the paper's central mechanism predicts is a small, self-consistent confirmation of it.

What does the correction actually do? Table 6 shows the correction works; it does not say what changes. We check directly by computing activation CKA between the corrected K and the (untouched) V on held-out data disjoint from whatever the map was fit on. Averaged over layers, the linear map raises CKA(K,V) from a baseline of 0.73 to 0.99 at synthetic/vision scale, and from 0.69 to 0.92 at LLM scale – essentially saturating the similarity measure. Procrustes, by contrast, moves it far less (0.73 → 0.76 small-scale, 0.69 → 0.72 at LLM scale), despite recovering comparable or better zero-shot accuracy in Table 6 (e.g. Procrustes beats the linear map outright on CIFAR-10 and CIFAR-100). Representational similarity and functional recovery are evidently not the same axis here: an orthogonal rotation barely nudges K toward looking like V in CKA terms, yet unlocks most of the same downstream performance. The unconstrained linear map's own singular values are informative about why it reaches such high CKA: their mean is well below 1 (synthetic/vision: 0.33; LLM: 0.016, with a similarly small standard deviation) rather than clustered near 1 as a near-rotation would be – the fit is not finding a change of basis so much as aggressively projecting onto and rescaling a small number of directions, more so at LLM scale than at synthetic/vision scale. This is consistent with the useful $K \to V$ relationship living in a comparatively low-dimensional subspace rather than requiring the map's full rank.

Does joint training's shared projection resemble K, V, or neither? A different question than anything above: what does the *from-scratch* `Q≠K=V` model's single learned c_{kv} actually look like, relative to the independently-trained teacher's K and V? A priori this could go either way – joint training might converge toward one of the two roles, or find a genuinely different synthesis neither resembles. At synthetic/vision scale the answer is unhelpfully mixed: across 9 tasks/datasets (18 layer-level comparisons), c_{kv} is closer to K in 7 and closer to V in 11, with no consistent pattern by task type or layer depth, and is sometimes *less* similar to either than K and V are to each other (e.g. REVERSE layer 1: teacher's own K–V CKA is 0.82, while c_{kv}'s CKA to each is only 0.45 and 0.48) – consistent with joint training finding its own synthesis rather than approximating either independently-trained projection. At LLM scale the picture is more consistent, and points the other way from what the KEEP_V-favoring results above might suggest: c_{kv} is closer to the teacher's K than to V in 16 of 20 layers (mean CKA 0.79 vs. 0.67). We read this as informative but not contradictory: a K-like representation works fine in the scratch model because every downstream weight co-adapted to it during training, while the identical representation is what fails catastrophically under KEEP_K precisely because nothing downstream is allowed to adapt – the paper's central distinction (post-hoc merging vs. joint constraint, Section 1) applies to *which representation ends up shared*, not just to the direction of transfer. We do not have an account for why the LLM shows a consistent bias where the small-scale models don't, and flag it as an open question rather than force a reading onto it.

4.4 K–V representational similarity as a diagnostic for collapse severity

Corollary 1 predicts that the severity of collapse should track how much the teacher's independently-trained K and V projections actually diverged. We measure this directly: for each teacher checkpoint, we capture $c_k(x)$ and $c_v(x)$ activations on held-out data via forward hooks and compute linear CKA [2] between the two activation matrices, averaged over layers. High CKA means K and V computed near-redundant functions (safe to tie); low CKA means they specialized (unsafe to tie).

Across all 9 tasks, Figure 4 shows the predicted negative relationship: SUB and COPY sit at the high-CKA (0.93–0.94) / zero-collapse corner, while REVERSE, SORT, SWAP, and every vision dataset cluster at moderate CKA (0.53–0.75) with real collapse. This global correlation is driven mainly by that between-group contrast – pointwise tasks let K and V converge to near-identical functions with no cost, while every routing-dependent task or dataset needs them to specialize at least somewhat.

How much should $n = 9$ and that between-group contrast worry us? A fair amount, and we report the numbers that say so rather than lead with the most favorable one. Pearson's $r = -0.84$ ($p = 0.005$) has a bootstrap 95% confidence interval of $[-0.98, -0.18]$ (10^4 resamples) – consistent with a strong relationship, but not tightly enough pinned down at $n = 9$ to rule out a much weaker one. Spearman's rank correlation, which does not reward the linear separation between

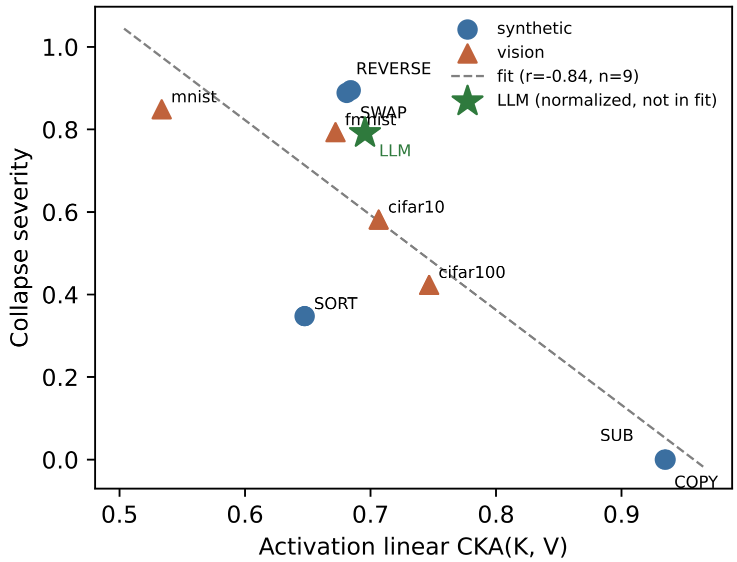

Figure 4: Activation-level linear CKA between a teacher's trained K and V projections vs. the accuracy drop under K=V surgery (averaged over KEEP_K and KEEP_V), across all 5 synthetic tasks and 4 vision datasets. Pearson $r = -0.84$ ($n = 9$); see text for why we do not treat this as a validated general predictor. The star marks the same measurement on the 300M-parameter FineWeb-Edu teacher (CKA $= 0.70$), with its y-coordinate a chance-normalized log-loss severity in place of accuracy drop (Table 4 only gives perplexity, which has no shared scale with accuracy): $(\ell_{\text{surgery}} - \ell_{\text{teacher}})/(\ln|V| - \ell_{\text{teacher}})$, where $\ell = \ln(\text{PPL})$ and $\ln|V|$ is the cross-entropy of a uniform predictor over the vocabulary – the LM analogue of chance accuracy. Excluded from the fit regardless, since this is a different derivation from the other 9 points' raw accuracy drop, not a directly validated equivalent measure.

the pointwise cluster and everything else as heavily as Pearson does, is $\rho = -0.56$ ($p = 0.12$, not significant). And within each task-type group alone – synthetic-only ($n = 5$: $r = -0.82$, $p = 0.09$) or vision-only ($n = 4$: $r = -0.85$, $p = 0.15$) – the correlation is directionally consistent with the global trend but does not reach significance at these small within-group sample sizes. We therefore do not treat CKA as a validated general predictor of collapse severity; it is a useful, cheap diagnostic that lines up with the theory's qualitative prediction across task types in this dataset, and the honest summary of the evidence for it is weaker than the headline $r = -0.84$ makes it look.

What the correlation does *not* explain is the ordering *within* the vision group. A natural hypothesis is that harder classification tasks force K and V to specialize *more* (lower CKA), which would explain why collapse gets worse from MNIST to CIFAR-100. The data says the opposite: activation CKA *rises* monotonically with dataset difficulty (MNIST 0.53 $\rightarrow$ FMNIST 0.67 $\rightarrow$ CIFAR-10 0.71 $\rightarrow$ CIFAR-100 0.75), even as zero-shot collapse also gets worse. So whatever drives increasing collapse severity across the vision datasets, it is not increasing K–V specialization – CKA is doing the opposite of what that story requires. We do not have an account for this and are not aware of one; we report it as an open question rather than force a narrative onto it. One plausible (but unverified) reading is that harder classification tasks lean more on the class token aggregating broad, low-precision context from many patches – a use of attention that Proposition 1 restricts regardless of how similar K and V are – so collapse severity here may be tracking how much content-routing the task depends on, not how specialized K and V happened to become. We leave testing that directly to future work.

Where in the network it happens. Every model here has exactly two attention layers, which lets us check layer-by-layer rather than only report the average. The two domains show opposite trends. In the three routing-dependent synthetic tasks, CKA is *lower* in layer 0 than layer 1 (REVERSE 0.54 $\to$ 0.82; SORT 0.56 $\to$ 0.74; SWAP 0.54 $\to$ 0.83): K and V specialize early and converge back toward redundancy by the final layer. In vision the trend reverses – CKA is high in layer 0 and drops in layer 1, the layer immediately feeding the CLS readout (MNIST 0.74 $\to$ 0.33; FMNIST 0.77 $\to$ 0.57; CIFAR-10 0.83 $\to$ 0.58; CIFAR-100 0.79 $\to$ 0.70). Layer 0's CKA across the four vision datasets is not monotonic in difficulty, but layer 1's is ($0.33 < 0.57 < 0.58 < 0.70$, i.e. MNIST $<$ FMNIST $\lesssim$ CIFAR-10 $<$ CIFAR-100): the across-dataset trend already visible in Figure 4 is carried almost entirely by the layer adjacent to the classification readout. This sharpens, without resolving, the open question above – whatever pushes harder vision datasets toward higher K–V redundancy concentrates at exactly the layer the class-token-aggregation account above would implicate, which is consistent with that account without being evidence for it over any other.

Robustness to the similarity measure. All CKA numbers above use the linear kernel. As a check that the pattern is not an artifact of that choice, we recomputed every measurement with RBF-kernel CKA [2] instead (median pairwise-distance bandwidth; activations subsampled to 2000 tokens per layer for tractability). The global correlation with collapse severity is essentially unchanged ($r = -0.87$ vs. $r = -0.84$ for linear CKA, $n = 9$, both $p < 0.01$), and the qualitative pattern is preserved throughout, including the vision layer-1 ordering above. We have not evaluated CCA- or mutual-information-based similarity measures; we have no reason to expect a different qualitative story from them, but we have not checked, and leave it to future work rather than claim it.

Weight-space cosine similarity between the raw (W_k, b_k) and (W_v, b_v) parameters, by contrast, is uninformative in every task (all values within ± 0.005 of zero) – as expected, since two linear maps can compute highly redundant *functions* while living in unrelated parameter bases; only the representation-level (CKA) comparison is basis-invariant and captures functional overlap.

5 Discussion

Recovery is fast relative to training from scratch: most (task, surgery-mode) configurations reach the from-scratch tied baseline within 1–5 epochs of distillation against the QKV teacher, versus the 10–50 epochs used to train the corresponding checkpoints from scratch, and KEEP_V/AVG on CIFAR are within a point of that baseline by 10 epochs (Table 2). One fact about the surgery is certain and worth stating plainly: it leaves every parameter *except* c_{kv} already correctly tuned (Section 1), so distillation is searching for a single linear map rather than re-deriving the whole network. Whether that alone accounts for the observed speed of recovery – as opposed to, say, the KL term against the teacher's soft logits doing more work than the search-space argument suggests – we have not verified, and we do not offer a convergence-rate account here. KEEP_K is the clear counter-case to any simple "small search space, therefore fast" story: it starts from the same reduced parameter count as KEEP_V and AVG but is consistently the slowest and least complete to recover (Table 2). Section 4.2's basis-mismatch account is more consistent with this than Corollary 1 as originally stated: distillation is not restoring content that KEEP_K discarded (a linear probe shows the content was never gone), it is re-calibrating c_{proj}/MLP/head to a representation in an unfamiliar basis – a harder fitting problem than the search-space argument alone would suggest. The same asymmetry survives the jump from the 1.6M-param CIFAR ViT to a 389M-parameter language model trained from scratch on 10B tokens – more than two orders of magnitude more parameters, and a far larger, qualitatively different pretraining corpus (Table 4): KEEP_K is still the worst zero-shot mode and the one left furthest from the tied ceiling after a fixed distillation budget, which weighs against reading the vision-scale KEEP_K gap as an artifact of the vision ViT being too small or CIFAR too little data for full re-calibration. The LLM's own content probe complicates this account rather than simply extending it, though (Section 4): unlike the synthetic/vision teachers, where K and V decoded token identity equally well, this teacher's K is measurably (if modestly) harder to decode than V at every layer – some genuine content asymmetry may be present at this scale on top of the basis-mismatch effect, not instead of it, and we do not have a clean way from the current evidence to apportion KEEP_K's slow recovery between the two.

The K→V alignment result (Section 4.3) sharpens what "re-calibrating to an unfamiliar basis" means concretely: a full knowledge-distillation pass updates every downstream parameter against a much richer training signal (hard labels plus the teacher's soft logits, many epochs or hundreds of millions of tokens), while the alignment correction updates *nothing* downstream and is a single closed-form linear or orthogonal fit on far less data. That such a cheap, purely representational fix gets most of the way to KEEP_V at every scale we tested – rather than distillation's slow, expensive re-calibration being necessary in principle – suggests the bulk of KEEP_K's deficit really is a change-of-basis problem, small compared to what generic downstream re-training would be needed for if the content itself were the missing piece. That distillation still converges slower for KEEP_K than for KEEP_V/AVG despite the fix being this cheap in principle is itself informative: gradient descent evidently does not find the closed-form correction directly, or finds it more slowly than it finds whatever KEEP_V/AVG need, which we have not investigated further.

6 Related work

Attention as constrained computation. Tsai et al. [4] reframe self-attention generally as a kernel smoother: y_i is a weighted average of value vectors, with the softmax scores acting as a data-dependent kernel. That framing already implies Proposition 1 as a special case once the vectors being smoothed and the vectors used to produce the kernel weights are forced to be the same object ($k = v$): the kernel-smoother view says the output lies in the convex hull of whatever is being smoothed, and our contribution is narrower and architecture-specific – showing what that hull becomes, and why it is a genuine bottleneck for some tasks and not others, exactly when $K = V$.

Two other lines of work probe different senses in which attention is "limited," and it is worth being precise about how ours differs from each. Dong et al. [5] show that stacked self-attention without skip connections or MLPs collapses token representations to rank 1 doubly exponentially with depth – a degeneracy that compounds across many layers of attention interacting with each other. Proposition 1's obstruction is unrelated to depth: it is a single-layer statement about the span reachable by one attention block once $K = V$, holds regardless of how many layers are stacked, and does not require removing the skip connections or MLPs that Dong et al. show arrest their collapse (our checkpoints keep both throughout). Raganato et al. [6] show that replacing most encoder attention heads with fixed, non-learned, position-based patterns barely hurts translation quality – i.e. the specific values of the attention weights a_{ij} are often not doing much specialized work. That is an empirical, training-time finding about how much freedom the *weights* need; ours is an architectural, weight-independent statement about what the *value space* can carry once $K = V$ – Proposition 1 holds for every possible a_{ij}, learned or fixed, so the limitation we describe is structural rather than something a better-trained or differently-patterned attention could route around.

Representational similarity. Our CKA measurements follow Kornblith et al. [2], and the finding that raw weight-cosine similarity is uninformative while activation-level CKA is not (Section 4) is the expected consequence of CKA's basis invariance – precisely the property it was designed to have.

Distillation. Our recovery protocol (hard-label cross-entropy plus KL divergence against soft teacher logits, $T = 2$) follows Hinton et al. [3]. That paper, and most of the compression literature built on it, uses distillation to shrink a model while preserving its architecture class; here distillation instead repairs a capacity loss introduced by an architectural edit made *after* training (Section 4) – a different use of the same mechanism.

KV-sharing architectures. $K{=}V$ tying is a special case of a broader, well-established family: multi-query attention shares K/V across all heads [7], and grouped-query attention shares them across groups of heads [8]; the base paper's own sweep includes exactly these variants (GQA/MQA and their `Q≠K=V` combinations) trained jointly from scratch at 300M and 1.2B scale. None of that is new, and we do not claim otherwise. What this note adds is orthogonal to the architecture choice itself: a mechanism for *why* imposing that same sharing structure *after* independent training is a qualitatively different operation from choosing it *before* training starts, and a cheap, testable account (Sections 2, 4.2) of when the difference is severe enough to matter. A reviewer's fair objection – "K/V-sharing is not new; this is already standard in efficient attention" – is correct about the architecture and, we think, orthogonal to what we are claiming about the post-hoc/joint-training distinction.

7 Conclusion

The result we would ask a reader to take away is not "$K = V$ is fundamentally deficient" – `Q≠K=V` trained jointly from scratch works fine (the base paper's headline result), and two of our five synthetic tasks are entirely untroubled by the surgery. The result is that **post-hoc merging of independently-trained representations is a different operation from a joint architectural constraint, with its own, separately characterizable failure mode**. Joint training lets addressing and content pressures negotiate a shared representation from the start; post-hoc merging imposes that constraint externally on two representations that specialized apart under no such pressure, and Proposition 1 gives a hard, narrow ceiling on what a single merged attention sub-block – not the whole network – can then carry (Remark 2).

Where that ceiling ends up mattering turned out to need more than the mechanism we started with. Corollary 1's original story – K addresses, V carries content, so K should be the less content-decodable of the two – does not survive a direct probe at synthetic/vision scale: K and V are equally linearly decodable there (Section 4.2). The same probe at LLM scale complicates rather than confirms this: K is measurably, if modestly, harder to decode than V at every layer of the 300M-parameter model (Section 4), closer to Corollary 1 as originally stated – so the content-loss and basis-mismatch accounts may both be partially right, at different scales or in different proportion, and we do not have the evidence to settle which. The evidence at small scale is more consistent with a more specific, basis-mismatch account than with Corollary 1 as originally stated. Measured against a random-projection baseline's own 5-seed spread rather than a single number, KEEP_K confers no detectable advantage over noise on most non-pointwise tasks and is dramatically worse on one (Table 3), even though its content is, by the same section's probe, fully present – and, more decisively, directly correcting K's basis toward V's with a learned linear or orthogonal map, fit on held-out data with no retraining of anything downstream, closes the entire KEEP_K gap and matches or exceeds KEEP_V at every scale we tested, including the 300M-parameter LLM (Section 4.3). That last result is a direct manipulation of the hypothesized mechanism, not an inference from indirect signals, and is the strongest evidence in the note for reading KEEP_K's failure as basis mismatch rather than content loss – though a learned correction working is consistent with the account without ruling out every alternative explanation of the same result, so we still stop short of calling it established. K–V activation CKA is a useful and cheap diagnostic that tracks collapse severity across task types in our data, but we do not oversell it as a validated general predictor: the correlation is weaker under rank statistics than under Pearson's, is not separately significant within either task-type group at our sample sizes, and runs in the opposite direction of naive expectation *within* the vision datasets for reasons we do not have an account for (Section 4) – three separate reasons for a future study with more than $n = 9$ tasks to revisit it. When the ceiling does matter, a short knowledge-distillation pass against the original model is a comparatively cheap and nearly-always-sufficient fix, with KEEP_K consistently the slowest to recover – which the basis-mismatch account predicts, since it is the mode with a genuinely unfamiliar representation to re-calibrate against, not merely a smaller one.

A LLM distillation curves

Table 4 reports only the zero-shot and final-budget perplexity for the LLM-scale confirmation (Section 4); Figure 5 shows the full recovery trajectory for all three surgery modes, in the same spirit as Figure 2 for the synthetic/vision runs.

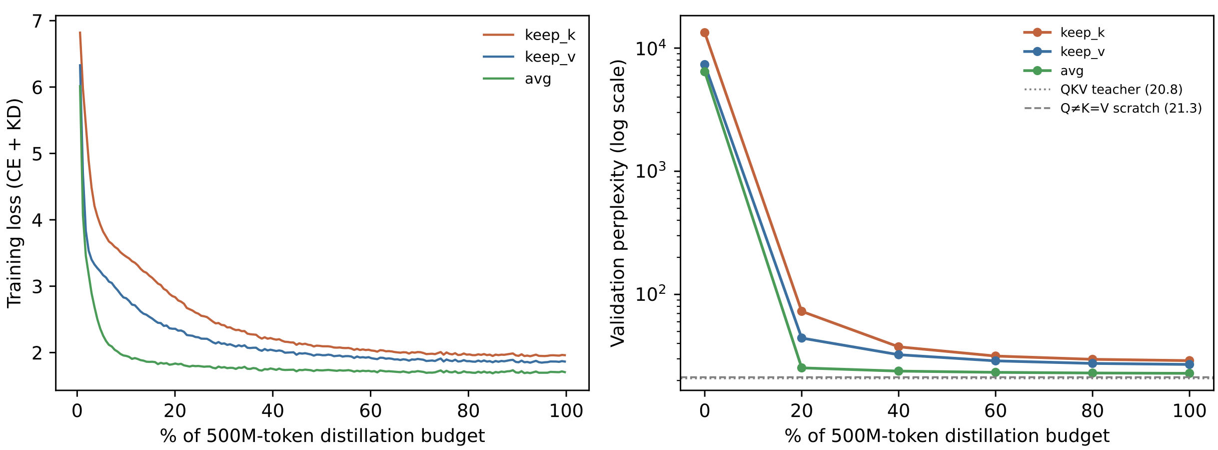

Figure 5: Left: training loss (hard-label CE + KD against the QKV teacher's soft logits) over the 500M-token distillation budget, parsed directly from the training logs (`distill_llm_{keep_k,keep_v,avg}.log`). Right: validation perplexity at the zero-shot point and each of the 5 evaluation checkpoints (log scale, given the roughly three-orders-of-magnitude range from zero-shot collapse to recovered perplexity), against the QKV teacher and from-scratch Q≠K=V ceiling (dotted/dashed reference lines). All three modes make most of their recovery in the first 20% of the budget and continue closing the gap to the ceiling gradually thereafter; AVG and KEEP_V track each other closely throughout, while KEEP_K remains visibly separated from both at every checkpoint, consistent with its persistent gap in Table 4 and the slower, less-complete KEEP_K recovery already seen at synthetic/vision scale (Figure 2).

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